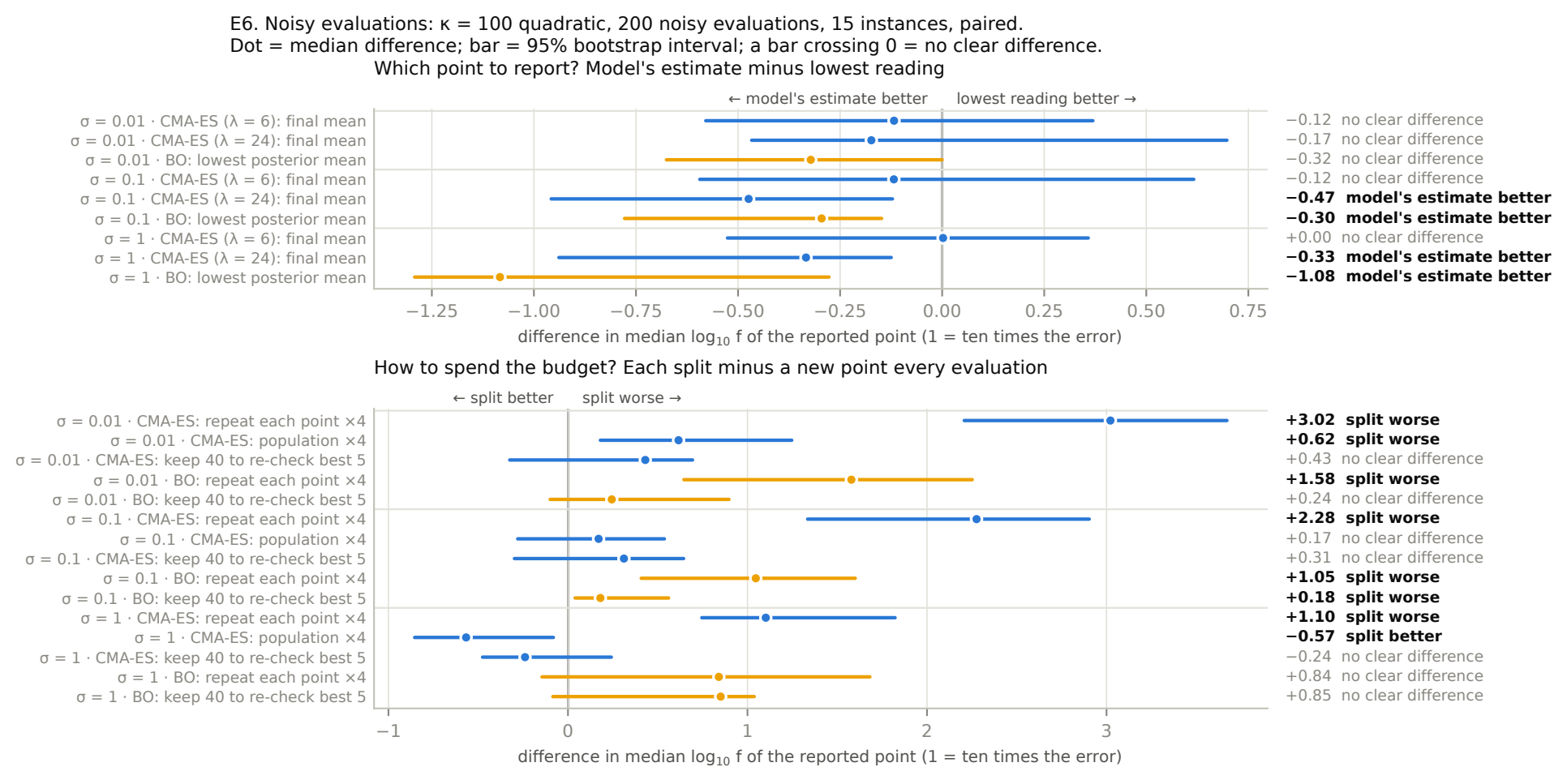
E6. Noisy evaluations: κ = 100 quadratic, 200 noisy evaluations, 15 instances, paired.
Dot = median difference; bar = 95% bootstrap interval; a bar crossing 0 = no clear difference.
Which point to report? Model's estimate minus lowest reading
← model's estimate better
lowest reading better →
σ = 0.01 · CMA-ES (λ = 6): final mean
σ = 0.01 · CMA-ES (λ = 24): final mean
σ = 0.01 · BO: lowest posterior mean
σ = 0.1 · CMA-ES (λ = 6): final mean
σ = 0.1 · CMA-ES (λ = 24): final mean
σ = 0.1 · BO: lowest posterior mean
σ = 1 · CMA-ES (λ = 6): final mean
σ = 1 · CMA-ES (λ = 24): final mean
σ = 1 · BO: lowest posterior mean
−0.12 no clear difference
−0.17 no clear difference
−0.32 no clear difference
−0.12 no clear difference
−0.47 model's estimate better
−0.30 model's estimate better
+0.00 no clear difference
−0.33 model's estimate better
−1.08 model's estimate better
−1.25 −1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75
difference in median log10 f of the reported point (1 = ten times the error)
How to spend the budget? Each split minus a new point every evaluation
← split better
split worse →
σ = 0.01 · CMA-ES: repeat each point ×4
σ = 0.01 · CMA-ES: population ×4
σ = 0.01 · CMA-ES: keep 40 to re-check best 5
σ = 0.01 · BO: repeat each point ×4
σ = 0.01 · BO: keep 40 to re-check best 5
σ = 0.1 · CMA-ES: repeat each point ×4
σ = 0.1 · CMA-ES: population ×4
σ = 0.1 · CMA-ES: keep 40 to re-check best 5
σ = 0.1 · BO: repeat each point ×4
σ = 0.1 · BO: keep 40 to re-check best 5
σ = 1 · CMA-ES: repeat each point ×4
σ = 1 · CMA-ES: population ×4
σ = 1 · CMA-ES: keep 40 to re-check best 5
σ = 1 · BO: repeat each point ×4
σ = 1 · BO: keep 40 to re-check best 5
+3.02 split worse
+0.62 split worse
+0.43 no clear difference
+1.58 split worse
+0.24 no clear difference
+2.28 split worse
+0.17 no clear difference
+0.31 no clear difference
+1.05 split worse
+0.18 split worse
+1.10 split worse
−0.57 split better
−0.24 no clear difference
+0.84 no clear difference
+0.85 no clear difference
−1 0 1 2 3
difference in median log10 f of the reported point (1 = ten times the error)